

$R \rightarrow$ orden del tensor

tensor de rotación

Ej 6

tensor
contracción

$$\rightarrow A_{\alpha_1 \dots \alpha_R} = B_{\alpha_1 \alpha_1} \dots B_{\alpha_R \alpha_R} A_{\alpha_1 \dots \alpha_R}$$

$$\alpha_1 = \alpha_2 \equiv j$$

$$\begin{pmatrix} B \end{pmatrix} \begin{pmatrix} B^T \end{pmatrix} = \begin{pmatrix} B^T \end{pmatrix} \begin{pmatrix} B \end{pmatrix} = \mathbb{I}$$

$$(\underline{B})_{ij} (\underline{B}^T)_{jk} = B_{ij} B_{kj} = \delta_{ik}$$

orden 2 orden 0
 $A_{ij} \rightarrow A_{ii}$

orden del tensor

$R-2$

dummy

$$\rightarrow A'_{jj \alpha_3 \dots \alpha_R} = \underbrace{B_{j \alpha_1} \cdot B_{j \alpha_2}}_{\delta_{\alpha_1 \alpha_2}} B_{\alpha_3 \alpha_3} \dots B_{\alpha_R \alpha_R} A_{\alpha_1 \alpha_2 \dots \alpha_R}$$

$$A'_{jj \alpha_3 \dots \alpha_R} = B_{\alpha_3 \alpha_3} \dots B_{\alpha_R \alpha_R} A_{\alpha_1 \alpha_1 \alpha_3 \dots \alpha_R}$$

$$\boxed{C'_{\pi_1 \dots \pi_{R-2}} = B_{\pi_1 \mu_1} \dots B_{\pi_{R-2} \mu_{R-2}} C_{\mu_1 \dots \mu_{R-2}}}$$

ley de transformación para un tensor de orden $R-2$